
Algorithm 1 Vector power method

Input: Matrix $A \in R^{n \times n}$

Output: (λ_1, v_1) eigenpair associated to the largest (in module) eigenvalue.

$v \in R^n$ given

$\beta = v \cdot A \cdot v$

repeat

$y = A \cdot v$

$v = y / \|y\|$

$\beta_{old} = \beta$

$\beta = v \cdot A \cdot v$

until $|\beta - \beta_{old}| / |\beta_{old}| < \varepsilon$

$\lambda_1 = \beta$ and $v_1 = v$
